

Python for Data Science - 2305CS303

Lab - 10

Roll No. : 128

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1. Series in Pandas

Create a Pandas Series containing the marks of 8 students. Display the Series, its index, values, and data type.

```
In [12]: import numpy as np
import pandas as pd

m = pd.Series([10,20,30],index=["raj","meet","jay"])
print(m)
```

```
raj      10
meet     20
jay      30
dtype: int64
```

2. Creating Series from Different Data

Create three Pandas Series: i) from a Python list, ii) from a NumPy array, and iii) from a dictionary. Display all three Series.

```
In [11]: list_data = [10, 20, 30, 40]
s_list = pd.Series(list_data)

array_data = np.array([1.5, 2.5, 3.5, 4.5])
s_array = pd.Series(array_data)

dict_data = {"Math": 88, "Science": 92, "English": 79}
s_dict = pd.Series(dict_data)

print("Series from List")
```

```
print(s_list)
print("\n Series from NumPy Array ")
print(s_array)

print("\n Series from Dictionary ")
print(s_dict)
```

Series from List

```
0    10
1    20
2    30
3    40
```

dtype: int64

Series from NumPy Array

```
0    1.5
1    2.5
2    3.5
3    4.5
```

dtype: float64

Series from Dictionary

```
Math      88
Science   92
English    79
```

dtype: int64

3. Custom Index in Series

Create a Pandas Series containing the marks of 5 students with student names as the index. Access the marks of two specific students using their labels.

In [15]: `marks = pd.Series([85, 92, 78, 90, 88], index=["raj", "sem", "jay", "nik", "keni`

```
print("--- Entire Series ---")
print(marks)

print()
selected_marks = marks[["kenil", "jay"]]
print(selected_marks)
```

--- Entire Series ---

```
raj      85
sem      92
jay      78
nik      90
kenil    88
```

dtype: int64

```
kenil    88
jay      78
```

dtype: int64

4. Creating Time Series

Create a date range for 10 consecutive days starting from '2026-09-01' and create a Pandas Series containing daily sales values for those dates. Display the resulting time series.

```
In [17]: dates = pd.date_range(start="2026-09-01", periods=10, freq="D")

sales_values = [150, 200, 180, 220, 210, 250, 300, 290, 310, 280]

sales_series = pd.Series(data=sales_values, index=dates)
print(sales_series)
```

```
2026-09-01    150
2026-09-02    200
2026-09-03    180
2026-09-04    220
2026-09-05    210
2026-09-06    250
2026-09-07    300
2026-09-08    290
2026-09-09    310
2026-09-10    280
Freq: D, dtype: int64
```

5. Time Series with Frequency

Create a time series for the first 6 months of 2026 using a suitable date frequency. Store monthly revenue values in the Series and display the result.

```
In [22]: months = pd.date_range(start="2026-01-01", periods=6, freq="MS")

revenue_values = [12000, 15000, 14000, 18000, 21000, 19500]
revenue_series = pd.Series(data=revenue_values, index=months)
print(revenue_series)
```

```
2026-01-01    12000
2026-02-01    15000
2026-03-01    14000
2026-04-01    18000
2026-05-01    21000
2026-06-01    19500
Freq: MS, dtype: int64
```

6. Creating a DataFrame

Create a DataFrame named `students` containing the following columns: RollNo, Name, Department, Marks, and City. Add records for at least 6 students. Display the DataFrame.

```
In [25]: data = {
    "RollNo": [101, 102, 103, 104, 105, 106],
    "Name": ["Aarav", "Bhavya", "Chetan", "Dev", "Esha", "Farhan"],
    "Department": ["CSE", "ECE", "CSE", "ME", "ECE", "CSE"],
    "Marks": [85, 92, 65, 88, 75, 58],
    "City": ["Mumbai", "Delhi", "Bangalore", "Chennai", "Hyderabad", "Pune"],
}

students = pd.DataFrame(data)
print("")
print(students)
```

	RollNo	Name	Department	Marks	City
0	101	Alice	CS	85	New York
1	102	Bob	ECE	92	Chicago
2	103	Charlie	CS	78	Boston
3	104	David	ME	88	Seattle
4	105	Emma	ECE	95	Austin
5	106	Frank	CS	82	Denver

7. DataFrame from Dictionary

Create a DataFrame from a dictionary containing information about 5 products:

ProductID, ProductName, Category, Price, and Quantity. Display the DataFrame and its columns.

```
In [36]: product_data = {
    "ProductID": [201, 202, 203, 204, 205],
    "ProductName": ["Laptop", "Mouse", "Keyboard", "Monitor", "Headphones"],
    "Category": [
        "Electronics",
        "Accessories",
        "Accessories",
        "Electronics",
        "Accessories",
    ],
    "Price": [1200, 25, 45, 300, 80],
    "Quantity": [10, 50, 40, 15, 30],
}

products_df = pd.DataFrame(product_data)
print(products_df)
print(products_df.columns)
```

	ProductID	ProductName	Category	Price	Quantity
0	201	Laptop	Electronics	1200	10
1	202	Mouse	Accessories	25	50
2	203	Keyboard	Accessories	45	40
3	204	Monitor	Electronics	300	15
4	205	Headphones	Accessories	80	30

Index(['ProductID', 'ProductName', 'Category', 'Price', 'Quantity'], dtype='object')

8. Conditional Selection

Using the `students` DataFrame, display: i) students having Marks greater than 70, ii) students belonging to the 'CSE' department, and iii) students having Marks between 60 and 80.

```
In [35]: students = pd.DataFrame(data)

marks_gt_70 = students[students["Marks"] > 70]

cse_students = students[students["Department"] == "CSE"]

marks_60_80 = students[(students["Marks"] >= 60) & (students["Marks"] <= 80)]
print(" Marks > 70")
print(marks_gt_70)
print("\n CSE Department ")
```

```
print(cse_students)
print("\n Marks Between 60 and 80")
print(marks_60_80)
```

Marks > 70

	RollNo	Name	Department	Marks	City
0	101	Aarav	CSE	85	Mumbai
1	102	Bhavya	ECE	92	Delhi
3	104	Dev	ME	88	Chennai
4	105	Esha	ECE	75	Hyderabad

CSE Department

	RollNo	Name	Department	Marks	City
0	101	Aarav	CSE	85	Mumbai
2	103	Chetan	CSE	65	Bangalore
5	106	Farhan	CSE	58	Pune

Marks Between 60 and 80

	RollNo	Name	Department	Marks	City
2	103	Chetan	CSE	65	Bangalore
4	105	Esha	ECE	75	Hyderabad

9. Multiple Conditions

Using the `students` DataFrame, select students who belong to the 'CSE' department and have Marks greater than or equal to 70. Also display students who belong to either 'CSE' or 'IT'.

```
In [30]: students = pd.DataFrame(data)

cse_and_high_marks = students[
    (students["Department"] == "CSE") & (students["Marks"] >= 70)
]

cse_or_it = students[
    (students["Department"] == "CSE") | (students["Department"] == "IT")
]

print("CSE department with Marks >= 70 ")
print(cse_and_high_marks)

print("\n belong to either CSE or IT ")
print(cse_or_it)
```

CSE department with Marks >= 70

	RollNo	Name	Department	Marks	City
0	101	Aarav	CSE	85	Mumbai

belong to either CSE or IT

	RollNo	Name	Department	Marks	City
0	101	Aarav	CSE	85	Mumbai
2	103	Chetan	CSE	65	Bangalore
5	106	Farhan	CSE	58	Pune

10. Conditional Selection on DataFrame Columns

From the `products` DataFrame, display products whose Price is greater than 500 and Quantity is less than 20. Then display only the ProductName and Price columns of the selected records.

```
In [34]: products_df = pd.DataFrame(product_data)

condition = (products_df["Price"] > 500) & (products_df["Quantity"] < 20)
filter_products = products_df[condition]

result = filter_products[["ProductName", "Price"]]

print("Selected Products (Price > 500 & Quantity < 20) ")
print(result)
```

```
Selected Products (Price > 500 & Quantity < 20)
  ProductName  Price
0      Laptop  1200
```

11. Read CSV File

Create a CSV file named `students.csv` containing student records. Read the CSV file using Pandas and display the complete DataFrame. Also display its first 5 records.

```
In [37]: df_to_save = pd.DataFrame(students)

df_to_save.to_csv("students.csv", index=False)
print("File successfully created \n")

df = pd.read_csv("students.csv")

print(df.head())
```

File successfully created

	RollNo	Name	Department	Marks	City
0	101	Aarav	CSE	85	Mumbai
1	102	Bhavya	ECE	92	Delhi
2	103	Chetan	CSE	65	Bangalore
3	104	Dev	ME	88	Chennai
4	105	Esha	ECE	75	Hyderabad

12. Read Excel File

Create an Excel file named `students.xlsx` containing student records. Read the Excel file using Pandas and display the DataFrame. Display the column names and shape of the DataFrame.

```
In [28]: df_write = pd.DataFrame(data)
df_write.to_excel("students.xlsx", index=False)
print("File created successfully \n")

df = pd.read_excel("students.xlsx")

print(df)
print(df.columns.tolist())
print(df.shape)
```

File created successfully

	RollNo	Name	Department	Marks	City
0	101	Aarav	CSE	85	Mumbai
1	102	Bhavya	ECE	92	Delhi
2	103	Chetan	CSE	65	Bangalore
3	104	Dev	ME	88	Chennai
4	105	Esha	ECE	75	Hyderabad
5	106	Farhan	CSE	58	Pune

```
['RollNo', 'Name', 'Department', 'Marks', 'City']  
(6, 5)
```

13. Setting Index

Using the `students` DataFrame, set the `RollNo` column as the index. Display the DataFrame and access the record of a student using the index label.

```
In [22]: data = {  
    "RollNo": [101, 102, 103, 104, 105, 106],  
    "Name": ["Aarav", "Bhavya", "Chetan", "Dev", "Esha", "Farhan"],  
    "Department": ["CSE", "ECE", "CSE", "ME", "ECE", "CSE"],  
    "Marks": [85, 92, 65, 88, 75, 58],  
    "City": ["Mumbai", "Delhi", "Bangalore", "Chennai", "Hyderabad", "Pune"],  
}  
  
students = pd.DataFrame(data)  
students.set_index('RollNo', inplace=True)  
print(students.loc[101])
```

```
Name      Aarav  
Department CSE  
Marks      85  
City      Mumbai  
Name: 101, dtype: object
```

14. Resetting Index

Reset the index of the DataFrame created in the previous question. Demonstrate both keeping the old index as a column and removing the old index.

```
In [25]: students.reset_index(inplace=True)  
  
print(students)
```

	RollNo	Name	Department	Marks	City
0	101	Aarav	CSE	85	Mumbai
1	102	Bhavya	ECE	92	Delhi
2	103	Chetan	CSE	65	Bangalore
3	104	Dev	ME	88	Chennai
4	105	Esha	ECE	75	Hyderabad
5	106	Farhan	CSE	58	Pune

15. Integrated Pandas Exercise

Create a student DataFrame containing RollNo, Name, Department, Marks, and City. Set RollNo as the index, select students with Marks ≥ 75 , reset the index, and finally save

the resulting DataFrame as both CSV and Excel files.

```
In [26]: students.set_index("RollNo", inplace=True)

result = students[students["Marks"] >= 75]

result.reset_index(inplace=True)
result.to_csv("students_result.csv", index=False)
result.to_excel("students_result.xlsx", index=False)

print(result)
```

	RollNo	Name	Department	Marks	City
0	101	Aarav	CSE	85	Mumbai
1	102	Bhavya	ECE	92	Delhi
2	104	Dev	ME	88	Chennai
3	105	Esha	ECE	75	Hyderabad

```
In [ ]:
```